SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	**Date:**	24-08-26

Code of Conduct

I __T Bhanu Teja/192472259___ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ___Bhanu Teja_____

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

1. Semantic Representation in Customer Support Chatbots.
   A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

ANSWER:

1. Analyze the semantic representations and identify the action–object relationships.

Semantic representation converts a natural-language customer query into a structured meaning representation containing the **action**, **object**, and **user/entity** involved.

Semantic analysis

Customer Query	Semantic Representation	Action	Object
Activate international roaming for my number.	ACTIVATE(Roaming, Customer)	ACTIVATE	Roaming
Deactivate caller tune service.	DEACTIVATE(CallerTune, Customer)	DEACTIVATE	CallerTune
Check my data balance.	QUERY(DataBalance, Customer)	QUERY	DataBalance
Enable 5G service.	ACTIVATE(5GService, Customer)	ACTIVATE	5GService

The action-object relationship can be represented as:

```
Customer
   |
   | performs
   ↓
 ACTION
   |
   ↓
 OBJECT / SERVICE
```

For example:

```
Customer
   |
   └── ACTIVATE ──→ Roaming


Customer
   |
   └── DEACTIVATE ─→ CallerTune


Customer
   |
   └── QUERY ────────→ DataBalance


Customer
   |
   └── ACTIVATE ──→ 5GService
```

Thus, semantic representation removes unnecessary grammatical variation and represents the **actual intention of the customer in a machine-readable form**.

2. Determine which query contains semantic interpretation errors.

The predicted and actual intents are:

Query	Actual Intent	Predicted Intent	Result
Q1	Activate Roaming	Activate Roaming	Correct
Q2	Deactivate Caller Tune	Activate Caller Tune	**Incorrect**
Q3	Check Data Balance	Query Data Balance	Correct
Q4	Enable 5G Service	Activate 5G Service	Correct

Error

Q2 contains the semantic interpretation error.

The actual query is:

"Deactivate caller tune service."

Its correct representation is:

DEACTIVATE(CallerTune, Customer)

However, the system predicts:

ACTIVATE(CallerTune, Customer)

The object **CallerTune** is correctly identified, but the action is reversed from **DEACTIVATE** to **ACTIVATE**.

Actual:

Customer → DEACTIVATE → CallerTune

≠

Predicted:

Customer → ACTIVATE → CallerTune

This is a significant semantic error because the chatbot may perform the **opposite operation** requested by the customer.

3. Evaluate how meaning representation affects chatbot decision accuracy.

Meaning representation directly affects the ability of a chatbot to convert a user's message into the correct action.

In this case, three out of four predicted intents match the actual intent.

Therefore, the given system achieves **75% intent accuracy** based on the provided labels.

The error in Q2 demonstrates that identifying only the object is not sufficient. The system must correctly understand the **action as well as the object**.

For example:

"Activate caller tune"

↓

ACTIVATE(CallerTune)

"Deactivate caller tune"

↓

DEACTIVATE(CallerTune)

A correct semantic representation helps the chatbot:

- identify customer intent;
- select the correct service operation;
- reduce ambiguity;
- generate appropriate responses;
- avoid performing an opposite or incorrect action.

Therefore, accurate meaning representation is essential for reliable telecom customer-support automation.

4. Recommend improvements to enhance semantic understanding.

The chatbot can be improved by using a **multi-stage semantic understanding pipeline**.

User Query

↓

Tokenization & POS Analysis

↓

Intent Detection

↓

Action Identification

↓

Object/Entity Identification

↓

Semantic Representation

↓

Context & Validation

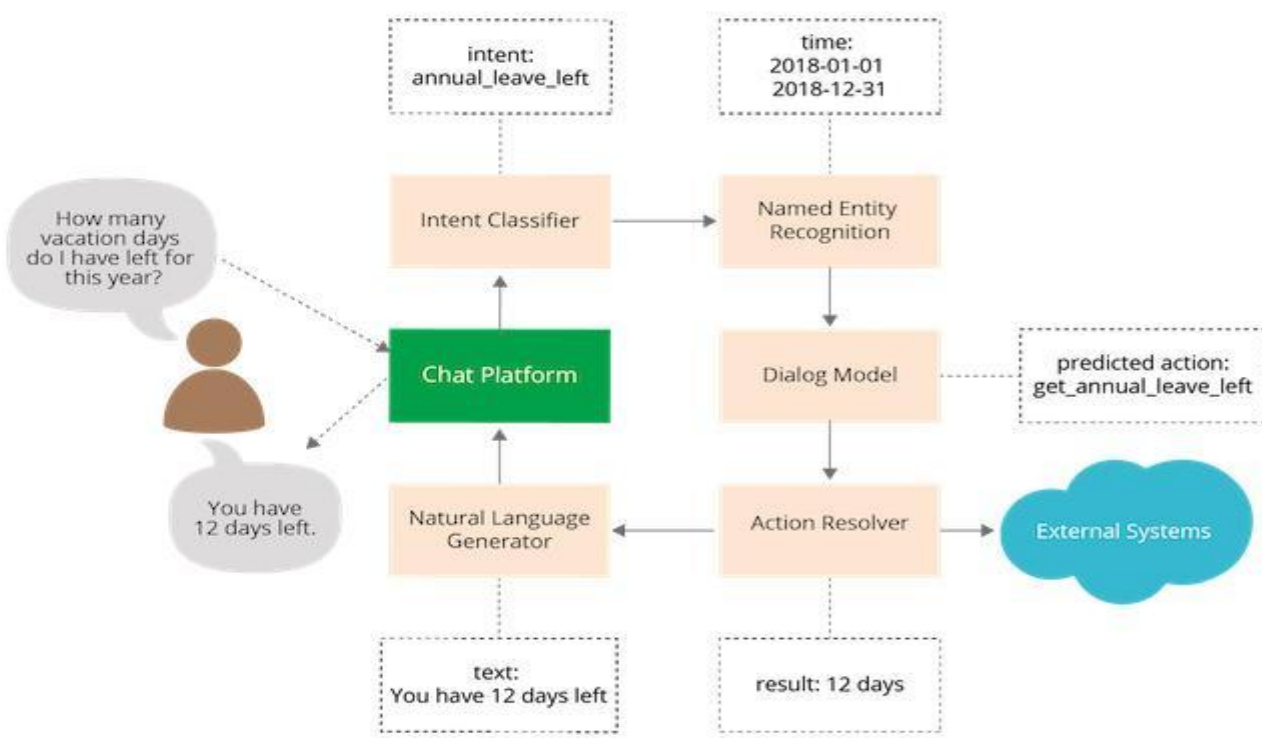
↓

Correct Action / Response

Recommended improvements

1. **Use context-aware intent classification** to distinguish words such as *activate*, *enable*, *deactivate*, and *disable*.
2. **Use synonym handling**, so that *enable* and *activate* can map to the same intent when appropriate.
3. **Use entity recognition** to accurately identify services such as roaming, caller tune, data balance, and 5G.
4. **Add semantic validation** before executing an action.
5. **Use confidence scores** and request clarification when the predicted intent has low confidence.
6. **Train using telecom-specific customer queries** containing different linguistic variations.

This would reduce semantic errors and improve the reliability of automated customer-support decisions.



2. First-Order Predicate Calculus for Smart Manufacturing.
An Industry 4.0 manufacturing plant monitors machines using semantic rules.
Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- Active(x) → Producing(x)
- Produces(x,y) ∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x)

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

ANSWER:

1. Represent the production data using First-Order Predicate Calculus.

The production data gives the status of four machines:

M1 → Active

M2 → Active

M3 → Maintenance

M4 → Active

The corresponding predicates are:

- Active(x) — machine x is active.
- Maintenance(x) — machine x is under maintenance.
- Producing(x) — machine x is producing.
- Produces(x,y) — machine x produces product y.
- Available(y) — product y is available.

Facts

Given rules

Rule 1:

This means every active machine is considered to be producing.

Rule 2:

This means if an active machine produces a product, that product is available.

Rule 3:

This means a machine under maintenance is not producing.

Logical representation

M1 ── Active ──→ Producing

M2 ── Active ──→ Producing

M3 ── Maintenance ──→ ¬Producing

M4 ── Active ──→ Producing

Therefore:

2. Analyze which products are currently available.

According to the rule:

a product is available when it is produced by an **active machine**.

For example, if:

and:

then:

Similarly, if:

and:

then:

Important observation

The case study does **not explicitly provide the Produces(x,y) facts** for the machines. Therefore, the exact product names cannot be logically determined from the given data alone.

The correct conclusion is:

Products produced by M1, M2, and M4 are currently available, provided the corresponding Produces(x,y) facts exist. Products dependent solely on M3 cannot be inferred as available because M3 is under maintenance.

Active Machines

M1 M2 M4

↓ ↓ ↓

Producing Producing Producing

↓ ↓ ↓

Products → AVAILABLE

M3

↓

Maintenance

↓

¬ Producing

3. Infer whether Gear production is affected by maintenance activities.

Maintenance directly affects production through:

Since:

we can infer:

If M3 is the machine responsible for Gear production, i.e.,

then Gear production is affected.

Therefore:

Maintenance(M3)

↓

¬Producing(M3)

↓

Gear production stops

↓

Gear may become unavailable

However, if Gear is produced by M1, M2, or M4, its production can continue because those machines are active.

Thus, the effect of maintenance depends on the **machine-product relationship**.

4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

First-Order Predicate Calculus is useful because it represents machine states and relationships formally and allows the system to derive new information from existing facts.

Example

Given:

and:

the system derives:

If:

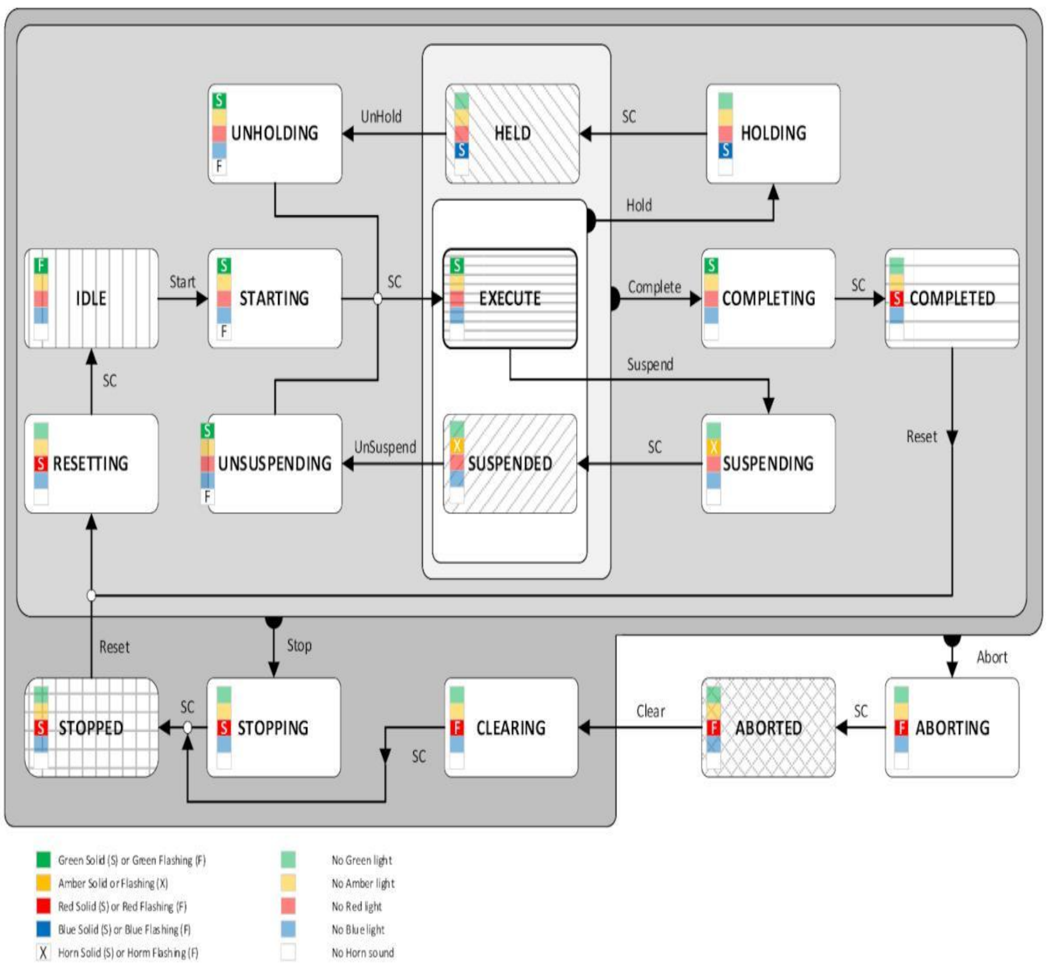
then:

Similarly:

Advantages

- Represents machine states precisely.
- Supports automated reasoning.
- Detects production constraints.
- Helps identify unavailable products.
- Supports maintenance-based decision making.
- Provides explainable logical conclusions.

Therefore, predicate logic provides a **structured and interpretable framework for industrial decision-support systems**, particularly where machine status and production dependencies must be reasoned about systematically.



3. Word Sense Disambiguation in E-Commerce Search Engines.
   An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand

"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	**Clicked Result**
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

ANSWER:

1. Analyze the contextual information and determine the correct word sense for each query.

Word Sense Disambiguation (WSD) identifies the correct meaning of an ambiguous word by considering the **context in which it occurs**.

User Query	Possible Senses	Context / Clicked Result	Correct Sense
Apple accessories	Fruit / Technology Brand	iPhone Charger	**Technology Brand**
Mouse wireless	Animal / Computer Device	Bluetooth Mouse	**Computer Device**
Java tutorial	Island / Programming Language	Coding Lessons	**Programming Language**
Python course	Snake / Programming Language	Software Development Training	**Programming Language**

Analysis

1. Apple accessories

The word *Apple* can refer to a fruit or the technology company. The clicked result is **iPhone Charger**, which is an electronic accessory. Therefore:

2. Mouse wireless

A wireless mouse in an e-commerce context normally refers to a **computer input device**, not an animal. The clicked result *Bluetooth Mouse* confirms this interpretation.

3. Java tutorial

Although Java can refer to an island, the word *tutorial* and the clicked result *Coding Lessons* indicate the programming language.

4. Python course

Python can refer to a snake or a programming language. The terms *course* and *Software Development Training* strongly indicate programming.

Overall WSD process

User Query

↓

Identify Ambiguous Word

↓

Analyze Surrounding Context

↓

Analyze Search/Click Information

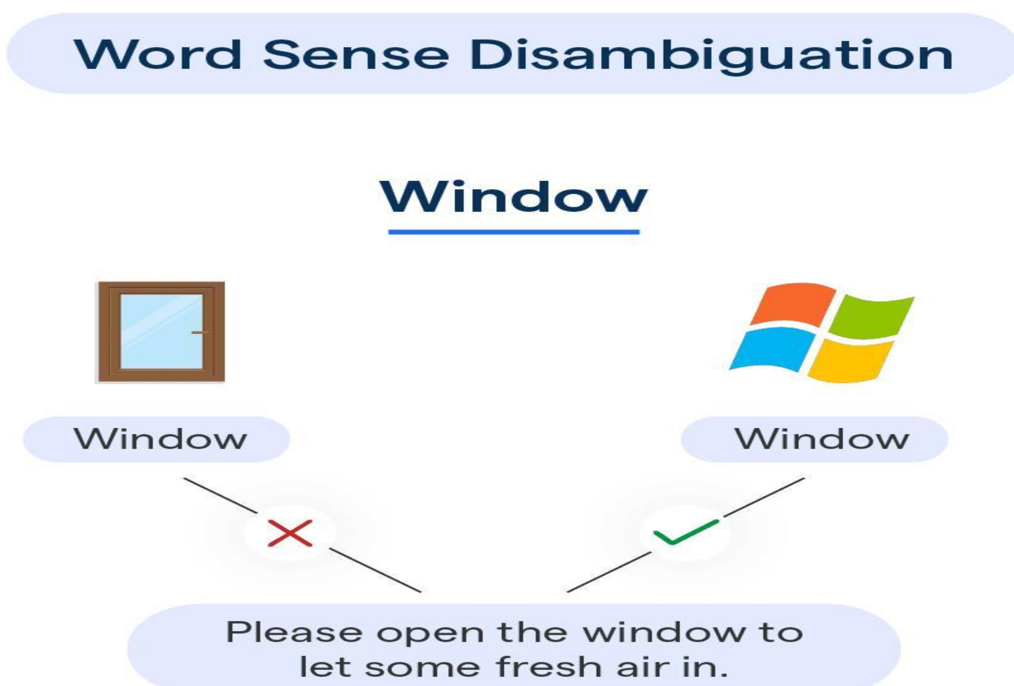
↓

Compare Possible Senses

↓

Select Most Probable Sense

Thus, **all four queries are correctly resolved using contextual and behavioral information**.



2. Identify the semantic cues used for disambiguation.

Several semantic and contextual cues help the search engine select the correct meaning.

Important cues

1. Neighboring words

Words occurring near the ambiguous term provide meaning.

Example:

Apple + accessories + iPhone Charger

↓

Technology Brand

2. Query-specific context

Words such as *tutorial*, *course*, and *wireless* restrict the possible meaning.

- *Java* + *tutorial* → Programming Language
- *Python* + *course* → Programming Language
- *Mouse* + *wireless* → Computer Device

3. Clicked search results

The clicked result provides strong evidence about the user's intended meaning.

For example:

"Apple accessories"

↓

iPhone Charger

↓

Technology Brand

4. Domain information

Since the system operates in an **e-commerce environment**, product-related interpretations receive higher importance.

5. User behavior

Search history, clicked products, previous queries, and purchasing behavior can also be used to improve WSD.

Therefore, WSD combines **linguistic context, semantic relationships, domain knowledge, and user behavior**.

3. Evaluate the impact of incorrect sense selection on search engine performance.

Incorrect WSD can significantly reduce the quality of an e-commerce search engine.

For example, if:

"Apple accessories"

is incorrectly interpreted as **Fruit**, the search engine might return fruit-related products instead of iPhone chargers and other Apple-device accessories.

Effects of incorrect WSD

- Irrelevant search results.
- Reduced recommendation accuracy.
- Lower user satisfaction.
- Increased search refinement attempts.
- Incorrect product ranking.
- Lower click-through rate.
- Possible loss of sales.

Example

User Query

"Apple accessories"

↓

Incorrect Sense: Fruit

↓

Fruit-related products

↓

Irrelevant Results

↓

Poor User Experience

Correct interpretation produces:

"Apple accessories"

↓

Technology Brand

↓

iPhone / Apple Accessories

↓

Relevant Results

↓

Higher User Satisfaction

Therefore, accurate WSD is important for **search relevance, recommendation quality, customer satisfaction, and business performance**.

4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

An industrial-scale WSD system should combine **NLP, machine learning, contextual information, and user behavior**.

Proposed architecture

User Query

↓

Query Processing

↓

Ambiguous Word Detection

↓

┌──────────────┼──────────────┐

↓ ↓ ↓

Linguistic Search User

Context History Behaviour

↓ ↓ ↓

└──────────────┼──────────────┘

↓

WSD Model

↓

Correct Word Sense

↓

Product Retrieval/Ranking

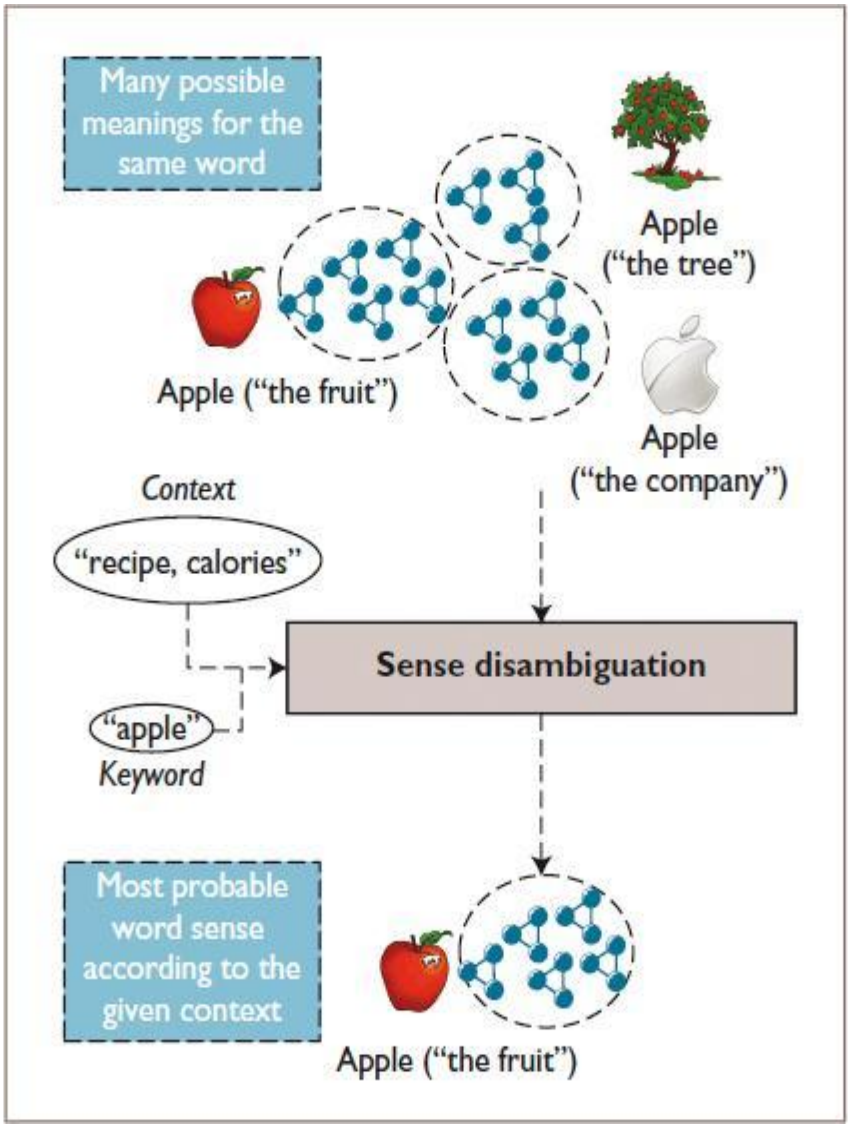
↓

Personalized Results

Recommended methods

1. **Context-aware language models** can understand the meaning of a word based on surrounding words.
2. **Entity recognition** can identify brands, products, locations, and technical terms.
3. **Click-through data** can be used as evidence of the user's intended sense.
4. **Search history** can provide additional context about the user's interests.
5. **Domain-specific knowledge bases** can distinguish meanings relevant to e-commerce.
6. **Confidence scoring** can identify uncertain cases and allow clarification.
7. **Continuous learning** can update the WSD model as new products, brands, and user queries appear.

Hence, combining **context + behavioral data + domain knowledge + machine learning** can provide scalable WSD and improve product recommendations.


Many possible meanings for the same word
Apple
("the tree")
Apple ("the fruit")
Apple
("the company")
Context
"recipe, calories"
Sense disambiguation
"apple"
Keyword
Most probable word sense according to the given context
Apple ("the fruit")

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

ANSWER:

1. Analyze how syntactic structures contribute to semantic role assignment.

Syntactic structure determines the relationship between different words in a sentence. The system uses this structure to identify **who performs an action, what is affected, and who receives it**.

For example:

Doctor prescribed medicine to patient.

The basic structure is:

Subject Verb Object Recipient

↓ ↓ ↓ ↓

Doctor prescribed medicine patient

↓ ↓ ↓ ↓

Agent Action Theme Recipient

Similarly:

Patient reported severe headache.

Patient ──→ reported ──→ severe headache

↓ ↓

Experiencer Symptom

Another example:

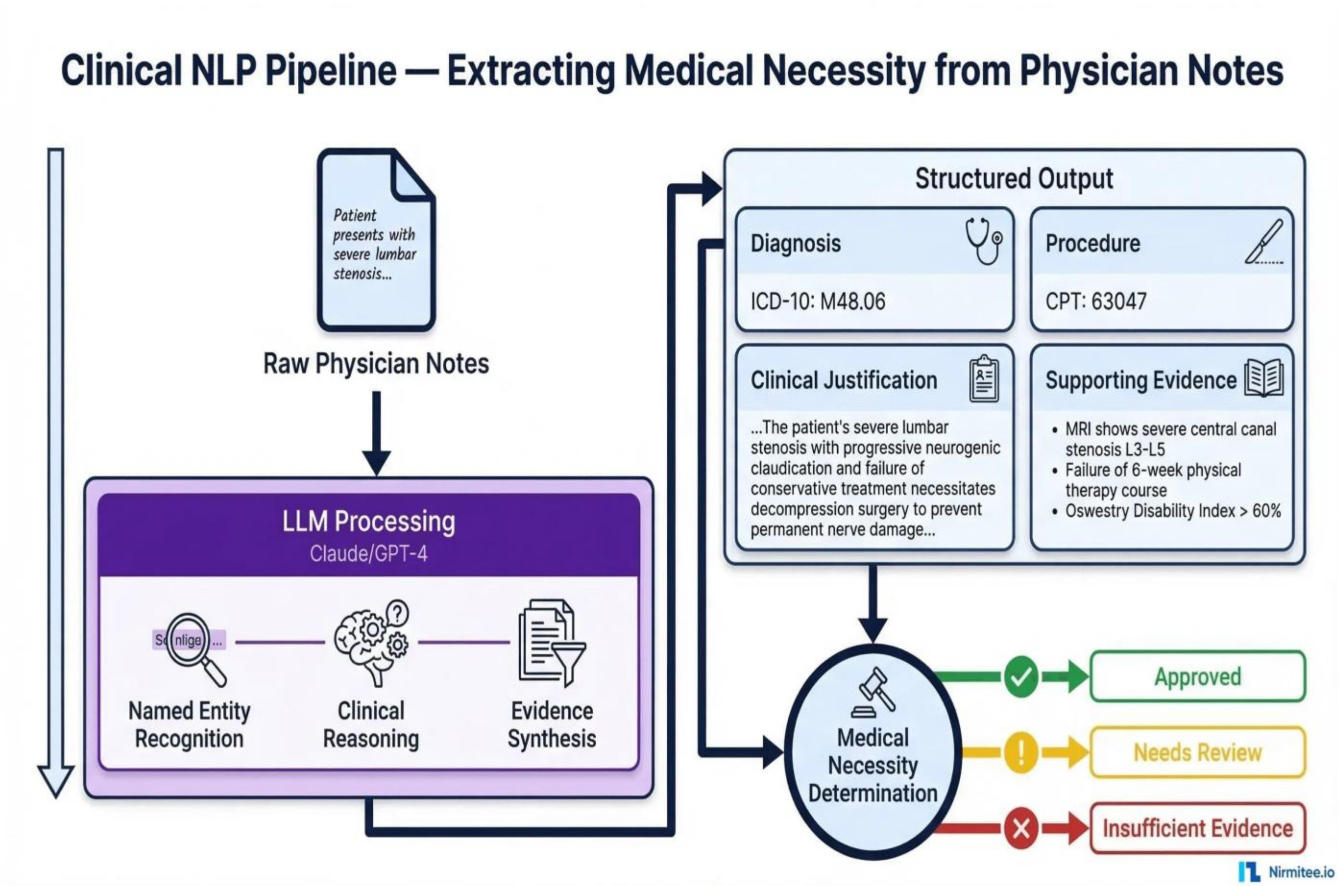
Nurse monitored patient continuously.

Nurse ──→ monitored ──→ patient

↓ ↓

Agent Patient/Theme

Thus, syntactic parsing helps the system identify **subject, verb, object, modifiers, and prepositional relationships**, which are then mapped to semantic roles.



2. Determine whether the assigned semantic roles are appropriate.

Some of the roles provided in the question are appropriate, but they need to be interpreted according to the **individual sentence context**.

Correct semantic-role analysis

Sentence	Entity	Appropriate Semantic Role
Doctor prescribed medicine to patient.	Doctor	**Agent**
	Medicine	**Theme**
	Patient	**Recipient**
Patient reported severe headache.	Patient	**Experiencer**
	Headache	**Symptom/Theme**
Nurse monitored patient continuously.	Nurse	**Agent**
	Patient	**Patient/Theme**
Medicine reduced blood pressure.	Medicine	**Cause/Instrument**
	Blood pressure	**Theme/affected entity**

Important correction

The given table lists **Medicine → Instrument**, which can be acceptable in some interpretations, but **Cause/Instrument** is more precise here because the medicine causes the reduction in blood pressure.

Also, **Patient → Recipient** is correct specifically for:

"Doctor prescribed medicine **to patient**."

But in:

"Patient reported severe headache."

the patient is better represented as an **Experiencer**, because the patient experiences/reports the symptom.

Therefore, semantic roles must be assigned using **both syntactic structure and sentence meaning**, rather than assigning one fixed role to a word across all sentences.

3. Evaluate potential errors that could arise from incorrect parsing.

Incorrect syntactic parsing can lead to incorrect semantic interpretation.

Example 1

Consider:

Doctor prescribed medicine to patient.

If the parser incorrectly identifies *patient* as the object receiving the action instead of *medicine*, the semantic representation may become incorrect.

Correct:

Doctor

↓ Agent

prescribed

↓

Medicine ——→ Theme

Patient ———→ Recipient

Possible errors

1. Wrong agent identification

The system may incorrectly identify the patient as the person performing the action.

2. Wrong semantic role

An object may incorrectly be classified as an agent, recipient, or instrument.

3. Incorrect medical information extraction

The system may misunderstand who received a medicine or who experienced a symptom.

4. Incorrect relation extraction

Relationships such as:

Doctor → prescribed → Medicine

Medicine → given to → Patient

may be incorrectly extracted.

5. Incorrect clinical decision support

Errors in semantic interpretation can affect medical summarization, information retrieval, and downstream decision-support systems.

Therefore, accurate syntactic parsing is particularly important in healthcare because semantic errors can change the interpretation of clinically relevant information.

4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

A robust healthcare NLP system should combine **syntactic parsing, semantic-role labeling, medical terminology, and contextual models**.

Recommended architecture

Medical Report

↓

Tokenization

↓

POS Tagging

↓

Syntactic Parsing

↓

Dependency / Constituency Structure

↓

Semantic Role Labeling

↓

Medical Entity Recognition

↓

Context & Negation Detection

↓

Clinical Knowledge Base

↓

Semantic Representation

Improvements

1. Use dependency parsing

Dependency relationships help identify subject, object, modifier, and other grammatical relationships.

2. Use semantic-role labeling

The system can explicitly identify roles such as:

- Agent
- Patient/Theme
- Recipient
- Experiencer
- Instrument
- Cause

3. Use medical-domain language models

Training or adapting NLP models using medical documents improves understanding of medical terminology.

4. Add negation detection

The system should distinguish:

"Patient has headache."

from:

"Patient does not have headache."

5. Use contextual analysis

The meaning of medical terms should be interpreted using surrounding words and the complete sentence.

6. Use medical knowledge bases

Medical ontologies and terminology databases can improve entity and relationship identification.

7. Apply confidence-based validation

Low-confidence semantic interpretations should be flagged for further validation rather than being directly used for critical decisions.

Final workflow

Syntax

↓

Grammar Relations

↓

Semantic Roles

↓

Medical Entities

↓

Context + Negation

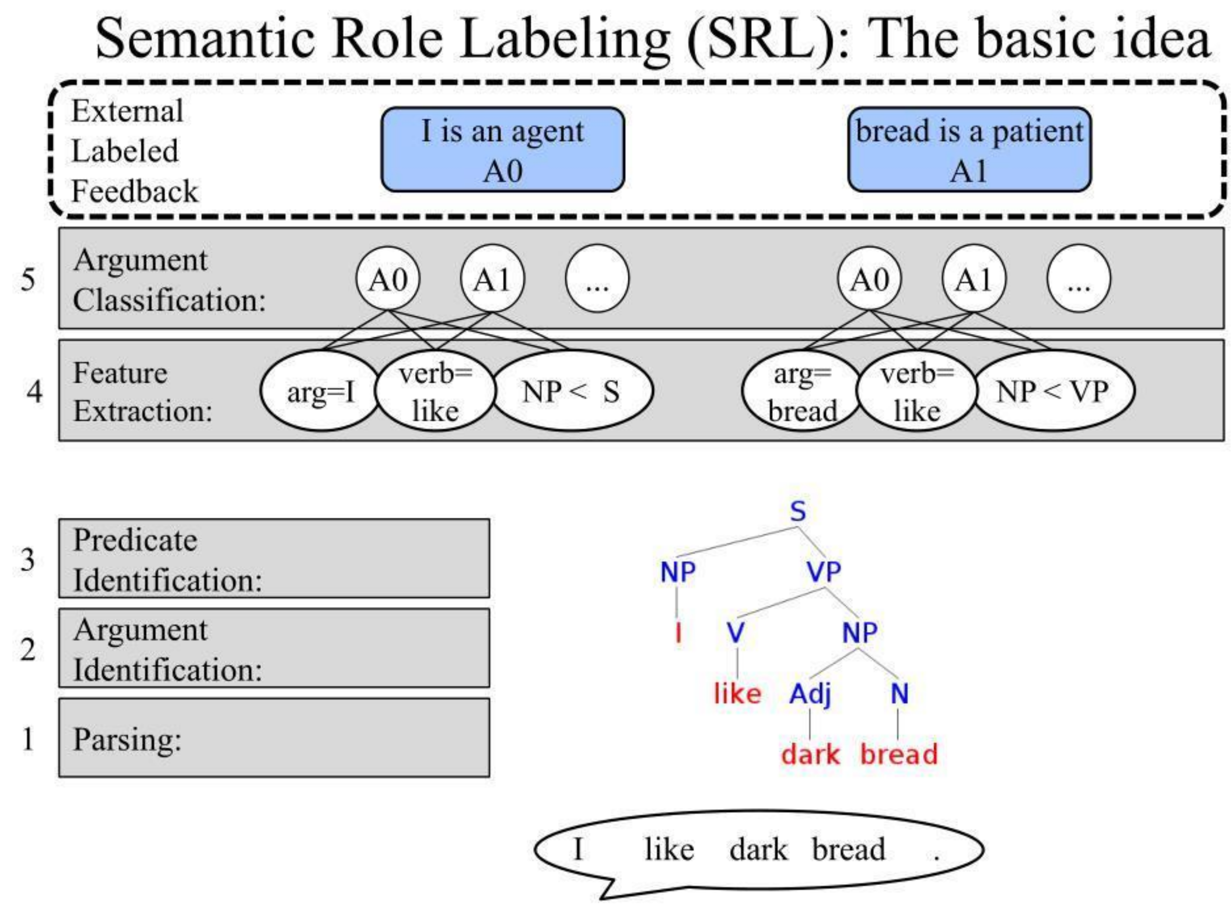
↓

Validated Clinical Meaning

↓

Conclusion

Syntax-driven semantic analysis provides the structural foundation for understanding medical text. By combining **syntactic parsing, semantic-role labeling, medical entity recognition, contextual analysis, and domain-specific knowledge**, healthcare NLP systems can extract more accurate clinical information and reduce semantic interpretation errors.



Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
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Date: ___________________	Date: ___________________	Date: ___________________